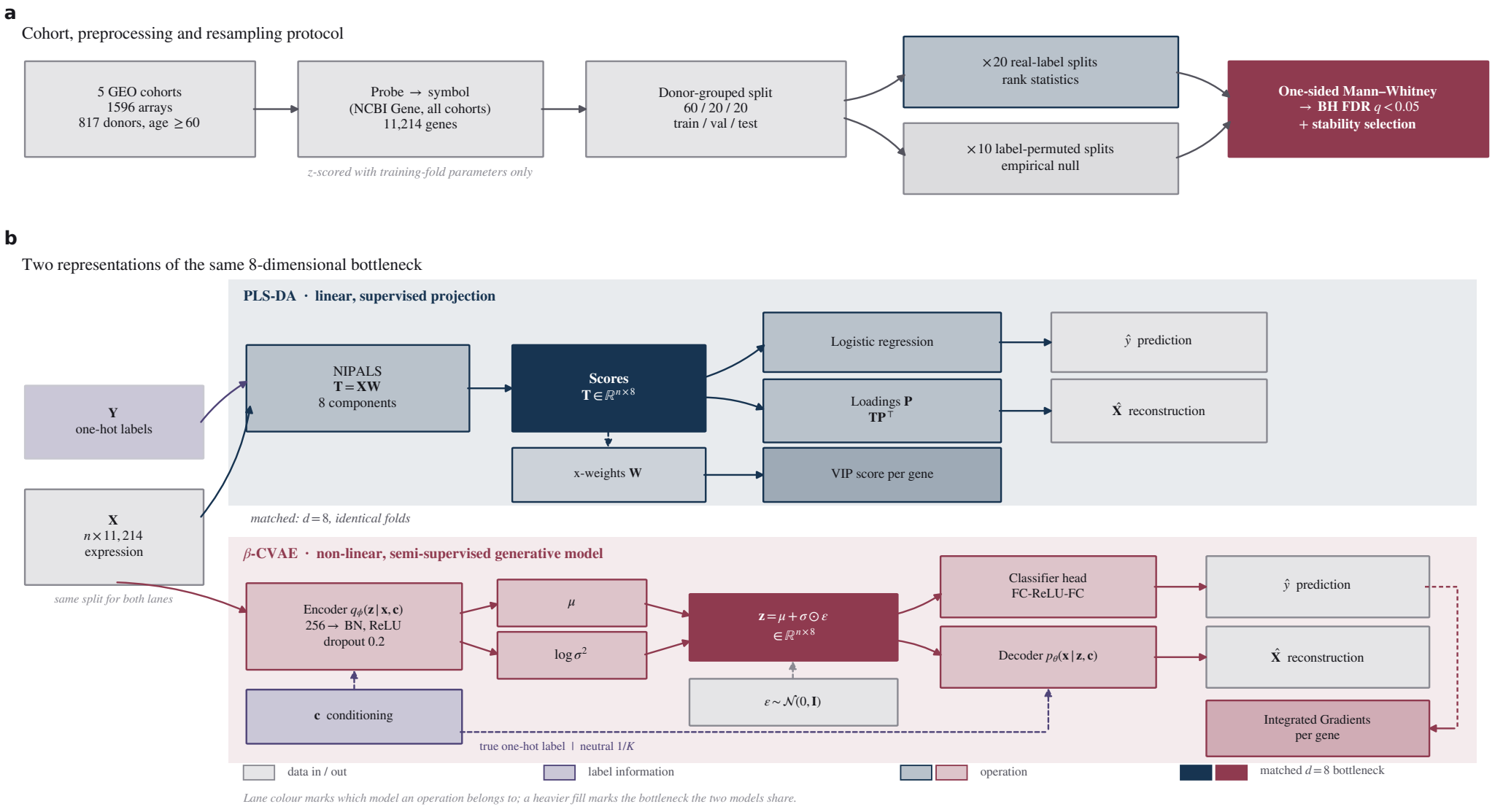
a
Cohort, preprocessing and resampling protocol
5 GEO cohorts
1596 arrays
817 donors, age ≥ 60
Probe → symbol
(NCBI Gene, all cohorts)
11,214 genes
z-scored with training-fold parameters only
Donor-grouped split
60 / 20 / 20
train / val / test
×20 real-label splits
rank statistics
×10 label-permuted splits
empirical null
One-sided Mann–Whitney
→ BH FDR q < 0.05
+ stability selection
b
Two representations of the same 8-dimensional bottleneck
PLS-DA · linear, supervised projection
Y
one-hot labels
X
n × 11,214
expression
same split for both lanes
NIPALS
T = XW
8 components
Scores
T ∈ ℝ^{n×8}
x-weights W
Logistic regression
Loadings P
TP^⊤
VIP score per gene
ŷ prediction
X̂ reconstruction
matched: d = 8, identical folds
β-CVAE · non-linear, semi-supervised generative model
Encoder q_φ(z | x, c)
256 → BN, ReLU
dropout 0.2
c conditioning
μ
log σ²
z = μ + σ ⊙ ε
∈ ℝ^{n×8}
ε ∼ 𝒩(0, I)
Classifier head
FC-ReLU-FC
Decoder p_θ(x | z, c)
ŷ prediction
X̂ reconstruction
Integrated Gradients
per gene
true one-hot label | neutral 1/K
data in / out
label information
operation
matched d = 8 bottleneck
Lane colour marks which model an operation belongs to; a heavier fill marks the bottleneck the two models share.